

Final: Review

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Omitted Variable Bias

For omitted variable bias to occur, the omitted variable “ Z ” must satisfy two conditions

- Z is a determinant of Y (i.e. Z is part of u); and
- Z is correlated with the regressor X (i.e. $\text{Corr}(Z, X) \neq 0$).

If there is omitted variable bias $\rho_X \neq 0$ and the *OLS* estimator $\hat{\beta}_1$ is biased and inconsistent.

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\sigma_u}{\sigma_X} \rho_{Xu} \quad (1)$$

Omitted Variable Bias

There are three ways to overcome the omitted variable bias

- 1 Run a randomized controlled experiment.
- 2 Adopt the “cross tabulation” approach.
- 3 Use a regression in which the omitted variable is no longer omitted (i.e., include the omitted variable when possible).

The Population Multiple Regression Model

Linear Regression Model With Multiple Regressors

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + u_i, \quad \forall i = 1, \dots, n. \quad (2)$$

- We have n observations $(Y_i, X_{1i}, \dots, X_{Ki})$.
- $X_1, \dots, X_K$ are the *independent* variables and Y is the *dependent* variable.
- β_0 is the intercept and $\beta_1, \dots, \beta_K$ are the slope parameters.
- u_i is the regression error, and it consists of omitted factors and errors in measurement of Y .

The Population Multiple Regression Model

Interpretation of the Estimated Slope Coefficients

Write the population model as

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_K X_K + u. \quad (3)$$

Consider changing X_1 by ΔX_1 while holding X_k constant $\forall k = 2, \dots, K$.
We can show

$$\beta_1 = \frac{\Delta Y}{\Delta X_1}, \text{ holding } X_k \text{ constant } \forall k = 2, \dots, K. \quad (4)$$

In general,

$$\beta_s = \frac{\Delta Y}{\Delta X_s}, \text{ holding } X_k \text{ constant } \forall k \neq s; k = 1, \dots, K. \quad (5)$$

OLS Estimator in The Multiple Regression Model

The OLS Estimator

Let $b_0, b_1, \dots, b_K$ be some estimators of $\beta_0, \beta_1, \dots, \beta_K$. Then, the predicted value of Y_i using this line is

$$\hat{Y}_i = b_0 + b_1 X_{1i} + \dots + b_K X_{Ki}.$$

Thus, the mistake made in predicting the i -th observation is

$$\begin{aligned}\hat{u}_i &= Y_i - \hat{Y}_i = Y_i - (b_0 + b_1 X_i + \dots + b_K X_{Ki}) \\ &= Y_i - b_0 - b_1 X_i - \dots - b_K X_{Ki},\end{aligned}$$

and the sum of the squared mistakes is

$$\sum_{i=1}^n (Y_i - b_0 - b_1 X_i - \dots - b_K X_{Ki})^2.$$

OLS Estimator in The Multiple Regression Model

The OLS Estimator

The *OLS* estimators of $\beta_0, \beta_1, \dots, \beta_K$ are the expressions for $b_0, b_1, \dots, b_K$ minimizing the sum of squared errors above. We will denote these estimators by $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_K$.

Measures of Fit for Multiple Regression

Recall

- $Y_i = \hat{Y}_i + \hat{u}_i$: actual = predicted + residual.
- SER : standard deviation of $\hat{u}_i$ (with d.f. correction).
- $RMSE$: standard deviation of $\hat{u}_i$ (without d.f. correction).
- R^2 : fraction of variance of Y explained by X .
- $\bar{R}^2$: Adjusted R^2 is the R^2 with d.f. correction, $\bar{R}^2 < R^2$.

Measures of Fit for Multiple Regression

R^2 and Adjusted R^2

- The “adjusted R^2 ” or $\bar{R}^2$ “penalizes” for the inclusion of additional regressors –the adjusted R^2 does not necessarily increase when you add another regressor (unlike the R^2).

$$\bar{R}^2 = 1 - \left(\frac{n-1}{n-K-1} \right) \frac{SSR}{TSS} = 1 - \left(\frac{n-1}{n-K-1} \right) (1 - R^2). \quad (6)$$

- Note that $\bar{R}^2 < R^2$ but their value is very close when n is large.

Units of SER, RMSE and R^2

- The units of the *SER*, *RMSE*, R^2 and $\bar{R}^2$ are the same as the dependent variable Y .
- *This is important to remember when comparing alternative models!*

OLS Assumptions for Multiple Regression

Recall our population model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + u_i, \quad \forall i = 1, \dots, n.$$

Least Square Assumptions

- ➊ **The conditional distribution of u_i given the X 's has mean zero.**
That is, $E[u_i | X_{1i} = x_1, \dots, X_{Ki} = x_K] = 0, \quad \forall i = 1, \dots, n.$
- ➋ **$(X_{1i}, \dots, X_{Ki}, Y_i), i = 1, \dots, n$, are i.i.d.**
- ➌ **Large outliers in X and/or Y are rare.** Technically,
 $0 < E[X_{1i}^4] < \infty, \dots, E[X_{Ki}^4] < \infty$, and $0 < E[Y_i^4] < \infty$
- ➍ **There is no perfect multicollinearity.**

Least Squares Assumptions

Assumption 4: There is no perfect multicollinearity

Perfect multicollinearity occurs when one of the regressors is an exact linear function of other regressors. This happens, for example, when you accidentally include the same variable twice.

Also, remember the dummy variable trap!

Sampling Distribution of the *OLS* Estimator

Under the four *OLS* assumptions above

- The sampling distribution of $\hat{\beta}_k$ has mean β_k .
- $\text{Var}[\hat{\beta}_k]$ is inversely proportional to n .
- Other than mean and variance the exact (finite sample) distribution of $\hat{\beta}_k$ is very complicated.
- For large n we can rely on the asymptotic approximation
 - $\hat{\beta}_k$ is consistent: $\hat{\beta}_k \xrightarrow{P} \beta_k$.
 - $\frac{\hat{\beta}_k - E[\hat{\beta}_k]}{\sqrt{\text{Var}[\hat{\beta}_k]}}$ is approximately distributed as a $N(0, 1)$ variable (by the *CLT*)
 - These statements hold for all variable k .

Perfect and Imperfect Multicollinearity

Perfect multicollinearity

Perfect multicollinearity is when one of the regressors is an exact linear function of the other regressors. In our *test score regression* some examples include

- you include *STR* twice, or dummy variable trap for example.

Imperfect Multicollinearity

Imperfect multicollinearity occurs when two or more regressors are very highly correlated.

- Imperfect multicollinearity implies that one or more of the regression coefficients will be imprecisely estimated.

Hypothesis Tests and Confidence Intervals for a Single Coefficient

- Hypothesis tests and confidence intervals for a single coefficient in multiple regression follow the same logic and recipe as for the slope coefficient in a single-regressor model
- Thus hypotheses on β_k can be tested using the usual *t-statistic*, and confidence intervals are constructed as before.

Tests of Joint Hypotheses

$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0 \text{ vs. } H_1 : \beta_1 \neq 0, \text{ or } \beta_2 \neq 0 \text{ or both.}$

- Individual *t-statistics* cannot be used to test a joint hypothesis unless you correct the significance level (Bonferroni method).
- The standard practice is to use an *F-statistic* to test both β_1 and β_2 at once.
- *Rule*: reject when F is large.
- The *p-value* is the tail probability of the χ^2_q/q (when comparing F) distribution beyond the *F-statistic* actually computed.
- Alternatively, the *p-value* is the tail probability of the χ^2 (when comparing $q \times F$) distribution.
- The “overall” regression *F-statistic* tests the joint hypothesis:
 $H_0 : \beta_1 = \dots = \beta_K = 0 \text{ vs. } H_1 : \beta_j \neq 0 \text{ for at least one } j.$

Tests of Joint Hypotheses

Homoskedasticity-only *F*-statistic

The homoskedasticity-only formula for the *F*-statistic is

$$F = \frac{(R^2_{unrestricted} - R^2_{restricted}) / q}{(1 - R^2_{unrestricted}) / (n - K_{unrestricted} - 1)}. \quad (7)$$

The bigger the difference between the restricted and unrestricted R^2 's—the greater the improvement in fit by adding the variables in question—the larger is the *homoskedasticity-only F*-statistic.

The *homoskedasticity-only F*-statistic rejects the null hypothesis when adding the two variables significantly increases the R^2 of the regression.

Testing Single Restrictions Involving Multiple Coefficients

For example to test the single restriction: $H_0 : \beta_1 = \beta_2$ vs. $H_1 : \beta_1 \neq \beta_2$.
There are three approaches

- 1 Perform the F -test directly: using software
- 2 Rearrange (“transform”) the regression
- 3 Perform a t -test with the appropriate standard error: you can extract the error manually from the estimated variance covariance matrix.

All these three methods are interrelated. In particular, method 2 is an easy way to get the standard error that you need without having to manually extract the required elements as in method 3. Both methods 2 and 3 perform the **same t -test!**

Confidence Sets for Multiple Coefficients

What is a joint confidence set for β_1 and β_2 ?

- A 95% joint confidence set is: the set of coefficient values that cannot be rejected at the 5% significance level. Equivalently, a set-valued function of the data that contains the true coefficient(s) in 95% of hypothetical repeated samples.
- Let $F(\beta_{1,0}, \beta_{2,0})$ be the (heteroskedasticity-robust) *F-statistic* testing the hypothesis $H_0 : \beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$. The 95% confidence set is $\{(\beta_{1,0}, \beta_{2,0}) : F(\beta_{1,0}, \beta_{2,0}) \leq 3.00\}$, where 3.00 is the 5% critical value of the $F_{2,\infty}$ distribution.

Model Specification for Multiple Regression

Control Variables

- A control variable W is a variable that is correlated with, and controls for, an omitted causal factor in the regression of Y on X , but which itself does not necessarily have a causal effect on Y .
- Control variables are **NOT** the object of interest; rather they are regressors we include to hold constant factors that, if neglected, could lead to the problem of omitted variable bias.

Model Specification for Multiple Regression

Conditional Mean Independence

When control variables are included, the *LSA (1)* $E[u_i|X_{1,i}, \dots, X_{K,i}] = 0$ must not hold. Control variables correlated with these omitted variables is precisely what makes it a good control variable!

Conditional mean independence: given the control variable, the mean of u_i does not depend on the variable of interest. Let X_i denote the variable of interest and W_i denote the control variable(s). Conditional mean independence holds if $E[u_i|X_i, W_i] = E[u_i|W_i]$.

Model Specification for Multiple Regression

Regression With Control Variables

Suppose that in addition to conditional mean independence the Least Square Assumptions (1) and (3) hold, then

- β_1 has a causal interpretation.
- $\hat{\beta}_1$ is unbiased.
- The estimator of the coefficient on the control variable, $\hat{\beta}_2$ is, in general, biased.
- This is not a problem for us because we do not aim to interpret the coefficient of the control variable.

A General Strategy for Modeling Nonlinear Regression Functions

General Nonlinear Population Model

The nonlinear population regression model is $Y_i = f(X_{1i}, \dots, X_{Ki}) + u_i$ for $i = 1, \dots, n$ where $f(X_{1i}, \dots, X_{Ki})$ is the population nonlinear regression function. We make the following assumptions

- 1 $E[u_i|X_{1i}, \dots, X_{Ki}] = 0$ (same as before); implies that f is the conditional expectation of Y given the X 's.
- 2 $(X_{1i}, \dots, X_{Ki}, Y_i)$ are i.i.d. (same).
- 3 Big outliers are rare (same idea; the precise mathematical condition depends on the specific f).
- 4 No perfect multicollinearity (same idea; the precise statement depends on the specific f).

A General Strategy for Modeling Nonlinear Regression Functions

- In linear models coefficients β_k have a natural interpretation.
- In nonlinear models interpretation of individual coefficients is very difficult and usually meaningless.
- We usually compute the predicted change in Y

$$\Delta \hat{Y} = \hat{f}(X_1 + \Delta X_1, X_2, \dots, X_K) - \hat{f}(X_1, X_2, \dots, X_K) \quad (8)$$

A General Strategy for Modeling Nonlinear Regression Functions

Standard Errors of the Estimated Effects

- Suppose we want to compute a confidence interval for the effect on Y , ΔY .
- It is easy to compute the standard error for $\Delta \hat{Y}$ when the regression is linear. For example, for a change in X_1 we have $\Delta \hat{Y} = \hat{\beta}_1 \Delta X_1$ and $SE(\Delta \hat{Y}) = SE(\hat{\beta}_1) \Delta X_1$.
- In the nonlinear model, $SE(\Delta \hat{Y}) = \frac{|\Delta \hat{Y}|}{\sqrt{F}}$, where F is the statistic testing an appropriate null hypothesis (see Ch.7).

Getting the standard error for an estimated effect.

Three approaches that parallel testing hypothesis involving a single restriction and multiple coefficients.

- 1 Perform the F -test directly: using software and compute
$$SE(\Delta \hat{Y}) = \frac{|\Delta \hat{Y}|}{\sqrt{F}}$$
- 2 Rearrange (“transform”) the regression to get the required SE
- 3 Compute manually the SE by extracting the items from the variance-covariance matrix.

Again, all these three methods are interrelated. In particular, method 2 is an easy way to get the standard error that you need without having to manually extract the required elements as in method 3. Method 1 is related to the other two because $F = t^2$.

Nonlinear Functions of a Single Independent Variable

Polynomials

Approximate the population regression function by a polynomial:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \dots + \beta_r X_i^r + u_i, \quad (9)$$

where r denotes the highest power of X . When $r = 2$ we have the quadratic regression model. Recall the procedure to select the degree of the polynomial.

- This is just the linear multiple regression model –except that the regressors are powers of X !
- Estimation, hypothesis testing, etc. proceeds as in the multiple regression model using *OLS*.
- The coefficients are difficult to interpret, but the regression function itself is interpretable.

Nonlinear Functions of a Single Independent Variable

- Another way to specify a nonlinear regression function is to use the natural logarithm of Y or X .
- Logarithms convert changes in variables into percentage changes. There are three different log regression specifications

	Case	Population Regression Model
I.	linear-log	$Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i$
II.	log-linear	$\ln(Y_i) = \beta_0 + \beta_1 X_i + u_i$
III.	log-log	$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i$

- The interpretation of the slope coefficient differs in each case.
- The interpretation is found by applying the general “before and after” rule: “figure out the change in Y for a given change in X .”
- Each case has a natural interpretation (for small changes in X).

Nonlinear Functions of a Single Independent Variable

Linear-log model

- Then, we have

$$Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i,$$

- and

$$\beta_1 \approx \frac{\Delta Y}{\Delta X/X}.$$

- Note that $100 \times \frac{\Delta X}{X}$ is the percentage change in X , so a 1% increase in X (this is multiplying X by 1.01) is associated with a $0.01\beta_1$ change in Y .
- Thus 1% increase in $X \rightarrow 0.01$ increase in $\ln(X) \rightarrow 0.01\beta_1$ increase in Y .

Nonlinear Functions of a Single Independent Variable

Log-linear model

- Then, we have

$$\ln(Y_i) = \beta_0 + \beta_1 X_i + u_i.$$

- and

$$\beta_1 \approx \frac{\Delta Y / Y}{\Delta X}.$$

- Now $100 \times \frac{\Delta Y}{Y}$ is the percentage change in Y , so an increase in X by one unit (this is $\Delta X = 1$) is associated with a $100\beta_1\%$ change in Y .

Nonlinear Functions of a Single Independent Variable

Log-log model

- Then, we have

$$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i.$$

- and

$$\beta_1 \approx \frac{\Delta Y / Y}{\Delta X / X}.$$

- Now $100 \times \frac{\Delta Y}{Y}$ is the percentage change in Y , and $100 \times \frac{\Delta X}{X}$ is the percentage change in X .
- Then, a 1% change in X is associated with a $\beta_1\%$ change in Y .
- In the *log-log* specification, β_1 has the interpretation of an elasticity.

Nonlinear Functions of a Single Independent Variable

Summary

- Three cases, differing in whether Y and/or X is transformed by taking logarithms.
- The regression is linear in the new variable(s) $\ln(Y)$ and/or $\ln(X)$, and the coefficients can be estimated by *OLS*.
- Hypothesis tests and confidence intervals are now implemented and interpreted “as usual.”
- The interpretation of β_1 differs from case to case.
- The choice of specification (functional form) should be guided by judgement (which interpretation makes the most sense in your application?), tests, and plotting predicted values

The end.